Predictive Resource Optimization Platform (PROP)

KPMG HEC CSD Data Challenge

PowerBI SQL implementation for data analysis

Introducing the Team

Team 7 - PROP

Minh Huynh
Software Engineering

Previous experience in software development and project management

Carter Cameron
Finance

Previous experience in mechanical & aerospace engineering

Casey Jussaume
Finance & Accounting

Previous experience in financial modelling at KPMG

Romero P. Faustin
Economics & Computer Science

Previous experience in data analysis

The Challenge & Mandate

Key business needs addressed

Charge-out rate variations

Timesheet & staffing data

Resource constraints

Objectives:

Maximizing profitability
Meet engagement deadlines
Minimize bench time

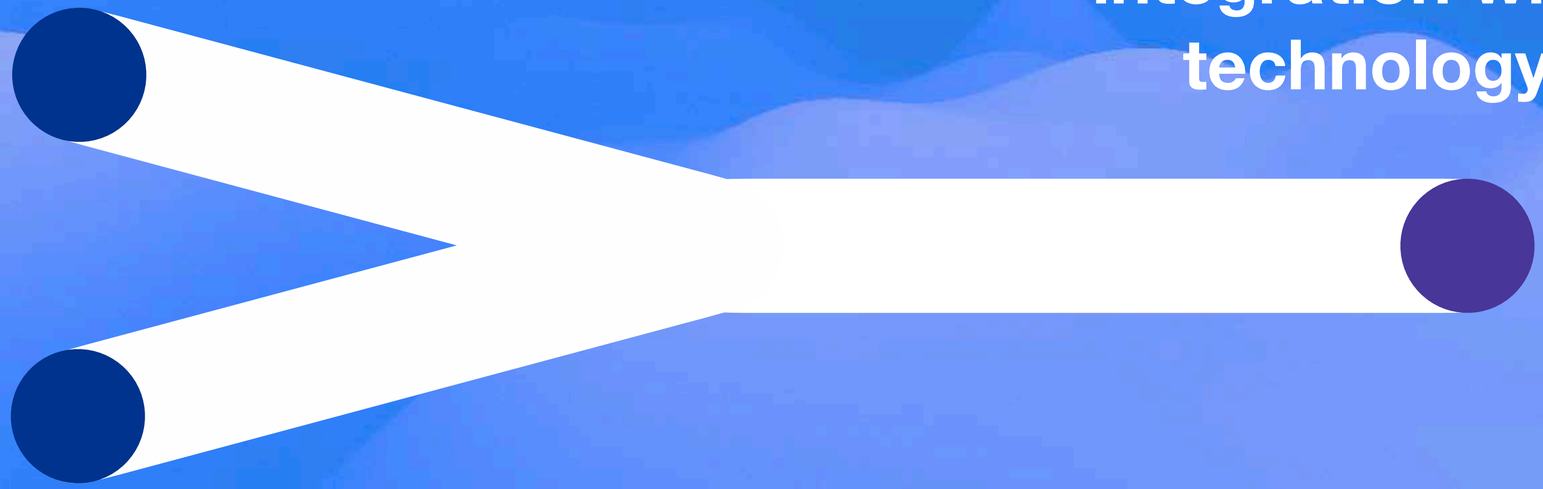
Project budgets

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Solution Overview

Introduction to PROP (Predictive Resource Optimization Platform)

Predictive modelling with optimization algorithm



Production-ready integration with KPMG technology stack

Visualizations through Power BI dashboards

Deliverables

Including an in-depth technical report

Optimization calculations in python

Source code

Power BI Dashboards

Presenting key insights using a direct connection to the SQL database

Database schema and setup scripts

DML and DDL SQL files

Microsoft® SQL Server®

Evolution from CPI to VEC

Introduction to PROP (Predictive Resource Optimization Platform)

CPI & EVM

Traditional EVM and CPI limitations for consulting

Only applicable to manufacturing – We need a metric that addresses the needs of consulting



VEC

How VEC addresses consulting's unique challenges

VEC captures financial efficiency and resource optimization

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Key Performance Indicators

Making informed business decisions

$$VEC_{ph} = \frac{BAC_{ph}}{\sum_{t=1}^{n} (AC_t \times d_t \times ea_c \times eo_c)}$$

$$Burn_{ph} = \frac{\sum AC_t}{BAC_{ph}}$$

Value Extraction Coefficient (VEC): Measuring financial realization efficiency

Budget Burn: Tracking cost control against budget

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Key Performance Indicators (cont'd)

Making informed business decisions

$$SPI_{ph} = \frac{EV_{ph}}{PV_{ph}}$$

$$\text{Benching Rate}_{c,w} = 1 - \frac{\sum_{en,ph} \text{staffing.planned_hours}_{c,en,ph,w}}{\text{employees.employment_basis}_{c}} \times 100\%$$

$$\text{Capacity Utilization Rate}_{c,w} = \frac{\sum_{en,ph} \text{timesheets.hours}_{c,en,ph,w}}{\text{employees.employment_basis}_{c}} \times 100\%$$

SPI:
Monitors delivery progress

Benching Rate:
Manages unutilized capacity

Capacity Utilization Rate:
Optimizes billable hours

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VEC Interpretation Table

Interpreting the VEC and Burn matrices

Meta	VEC > Burn Premium Realization	VEC ≈ 1	VEC < Burn (Value dilution)
Burn < 1	Excellent	On track	Warning
Burn ≈ 1	Excellent	On track	Warning
Burn > 1	Ambiguous	Critical	Critical

Algorithm Optimization

Maximizing weighted total engagement value & minimizing penalties

$$\max_{A} \sum_{en \in EN} w_{en} \times r_{en} \times (\alpha \times SPI_{en} + \beta \times VEC_{en}) - \sum_{c,en,w} pse \times is_switch_{c,en,w} - \sum_{en,l,pr,w} Penalty_{l,pr,en,w}$$

1. **Penalize Switching Costs:**

$$\sum_{c,en,w} pse \times is_switch_{c,en,w}$$

Where:

- pse — Phase Switching Efficiency penalty, applied when a consultant switches engagements.
- $is_switch_{c,en,w}$ — Binary variable indicating if consultant c switches to engagement en in week w.

Goal: Discourage frequent switching of consultants to avoid productivity losses.

2. **Penalize Deviations from Staffing Patterns:**

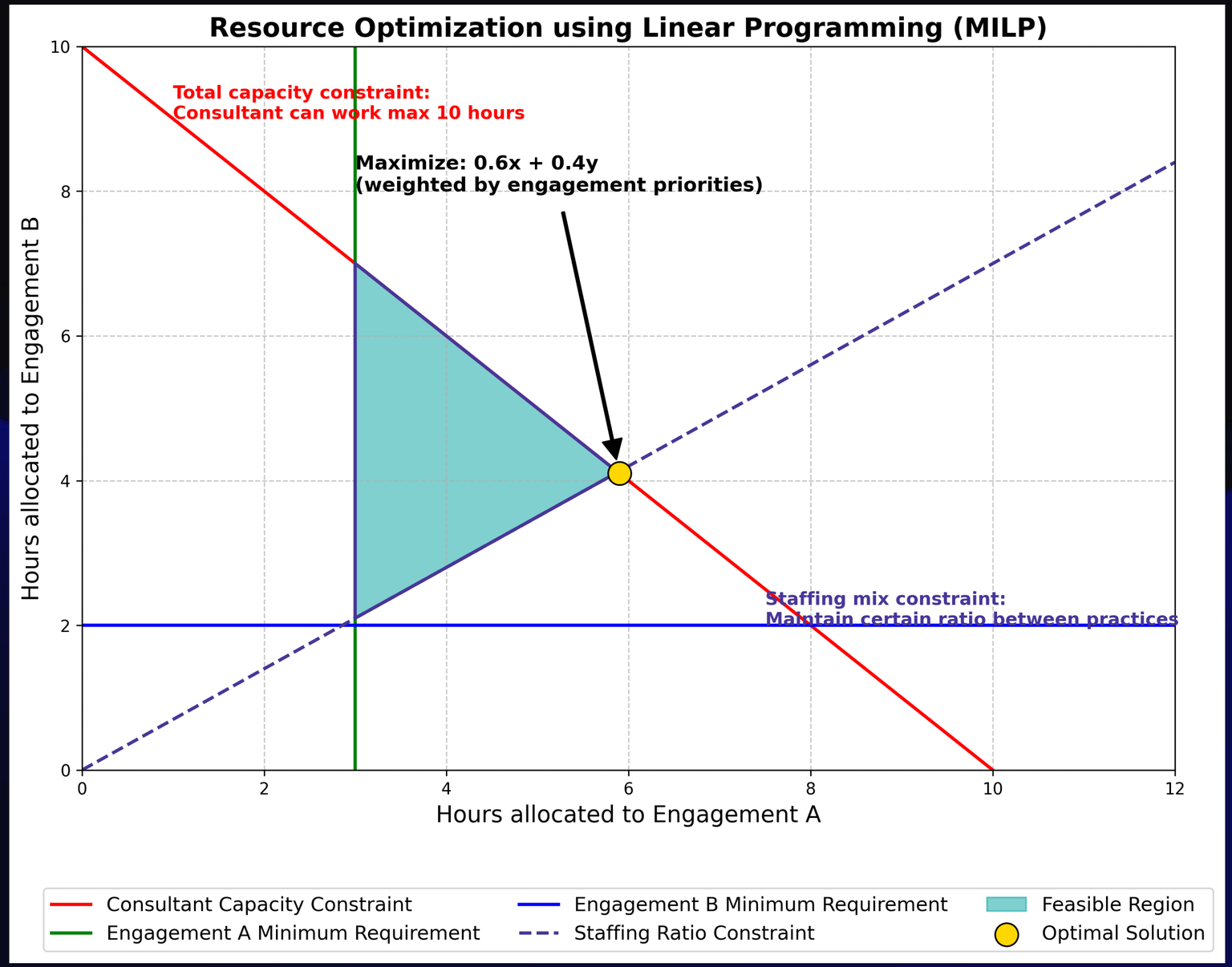
$$\sum_{en,l,pr,w} Penalty_{l,pr,en,w}$$

Where:

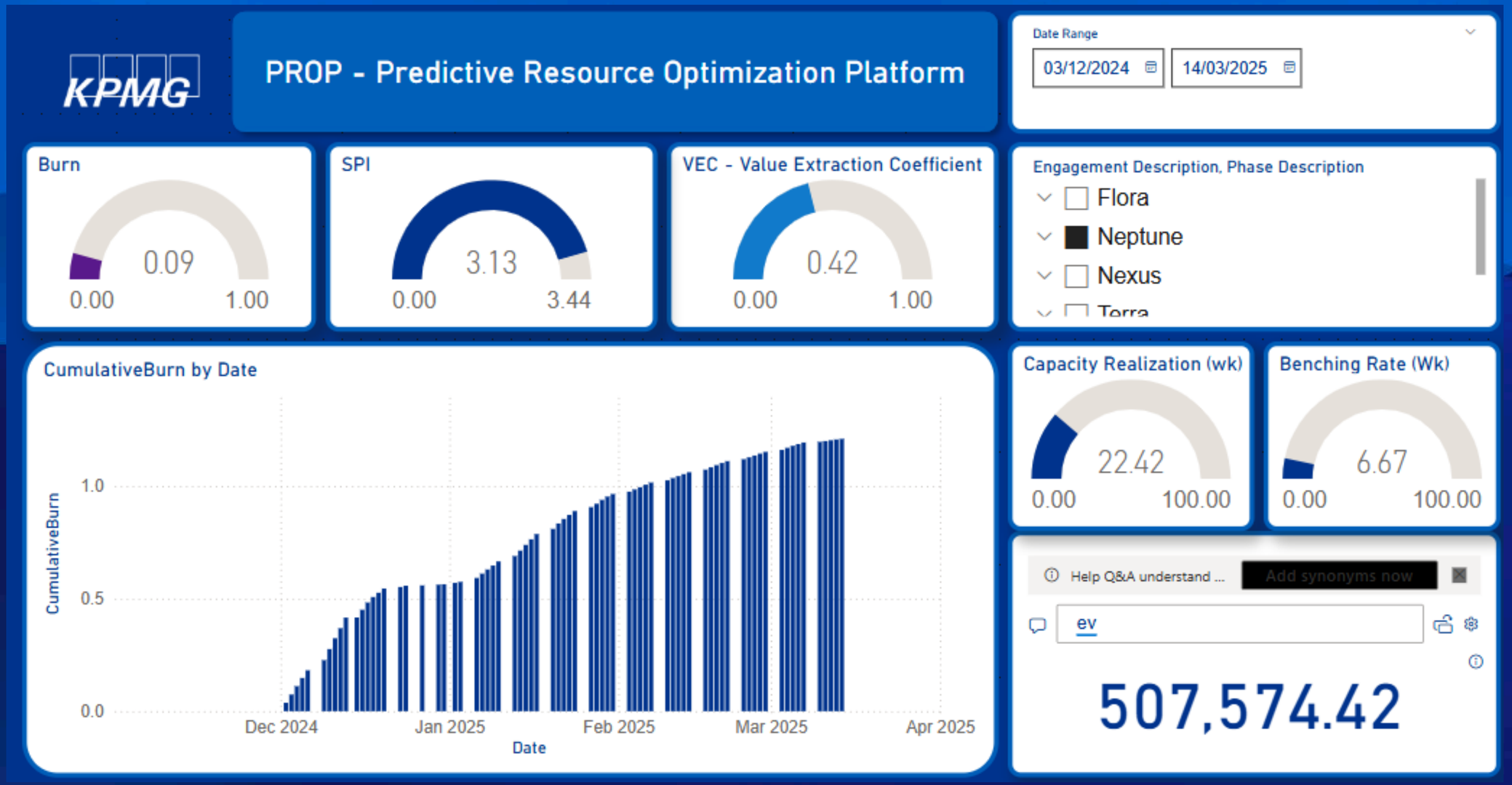
- $Penalty_{l,pr,en,w}$ — Penalty for deviations from ideal staffing ratios.
- l — Staff level (e.g., Manager, Consultant).
- pr — Practice area.

Goal: Ensure consistency with historical staffing patterns, reducing disruptions from significant staffing changes.

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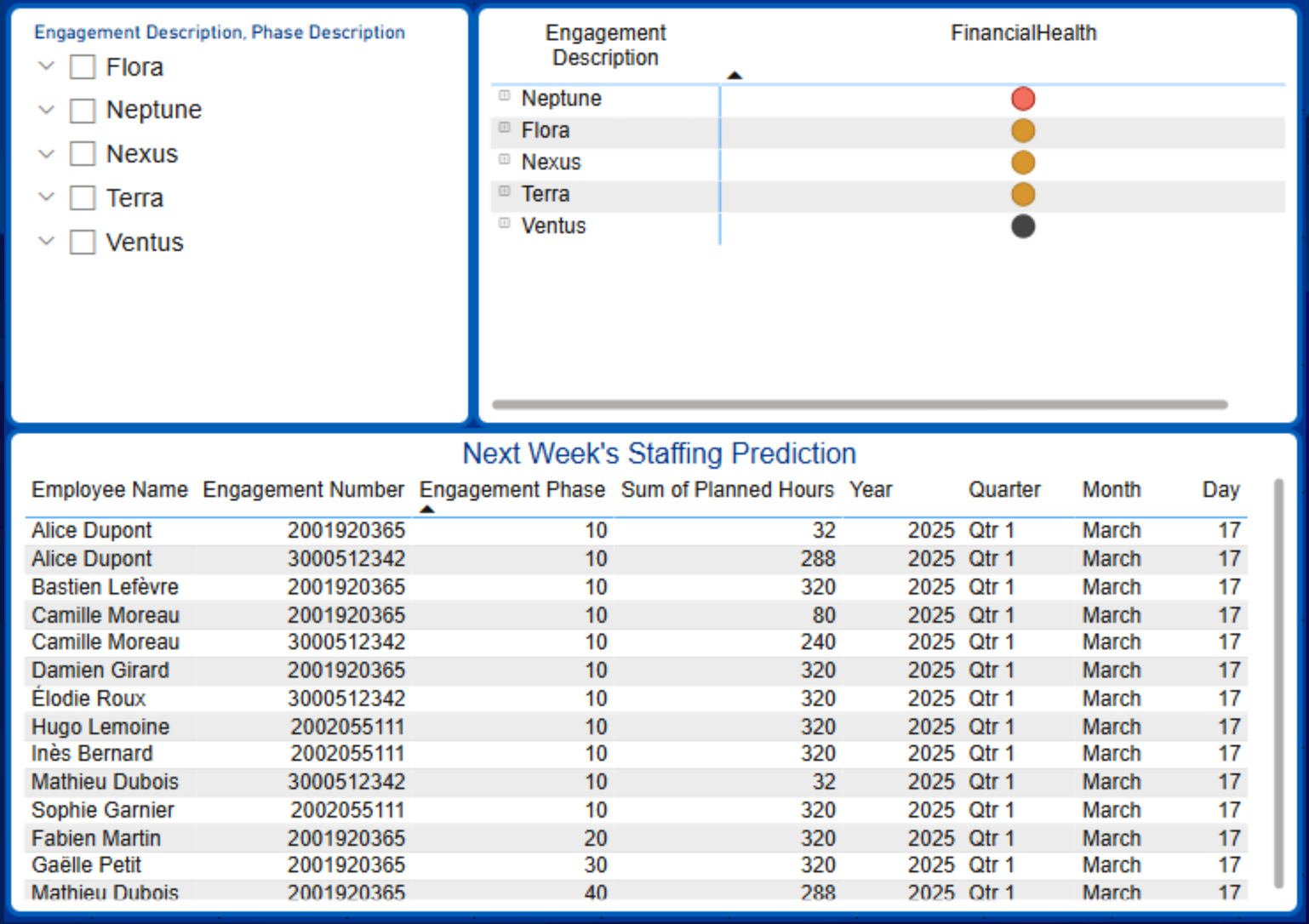

Resource Optimization using Linear Programming (MILP)
Total capacity constraint:
Consultant can work max 10 hours
Maximize: 0.6x + 0.4y
(weighted by engagement priorities)
Staffing mix constraint:
Maintain certain ratio between practices
Hours allocated to Engagement B
Hours allocated to Engagement A
0
2
4
6
8
10
12
Consultant Capacity Constraint
Engagement A Minimum Requirement
Engagement B Minimum Requirement
Staffing Ratio Constraint
Feasible Region
Optimal Solution

Primary Dashboard



Predictive Dashboard

Key inputs and interpretation of results



Next Week's Staffing Prediction

Employee Name	Engagement Number	Engagement Phase	Sum of Planned Hours	Year	Quarter	Month	Day
Alice Dupont	2001920365	10	32	2025	Qtr 1	March	17
Alice Dupont	3000512342	10	288	2025	Qtr 1	March	17
Bastien Lefèvre	2001920365	10	320	2025	Qtr 1	March	17
Camille Moreau	2001920365	10	80	2025	Qtr 1	March	17
Camille Moreau	3000512342	10	240	2025	Qtr 1	March	17
Damien Girard	2001920365	10	320	2025	Qtr 1	March	17
Élodie Roux	3000512342	10	320	2025	Qtr 1	March	17
Hugo Lemoine	2002055111	10	320	2025	Qtr 1	March	17
Inès Bernard	2002055111	10	320	2025	Qtr 1	March	17
Mathieu Dubois	3000512342	10	32	2025	Qtr 1	March	17
Sophie Garnier	2002055111	10	320	2025	Qtr 1	March	17
Fabien Martin	2001920365	20	320	2025	Qtr 1	March	17
Gaëlle Petit	2001920365	30	320	2025	Qtr 1	March	17
Mathieu Dubois	2001920365	40	288	2025	Qtr 1	March	17

Optimization recommendations

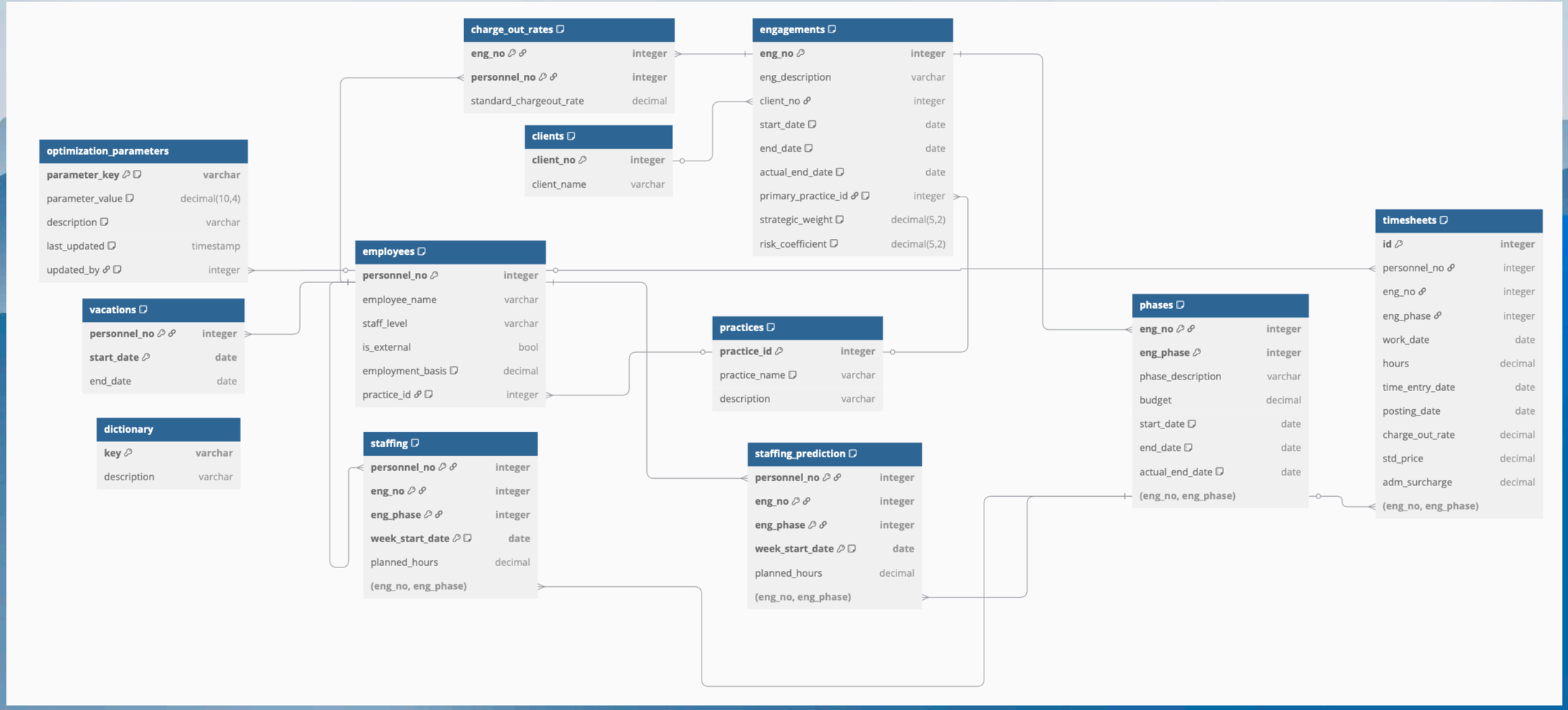
Resource allocation predictions

Potential conflicts or issues

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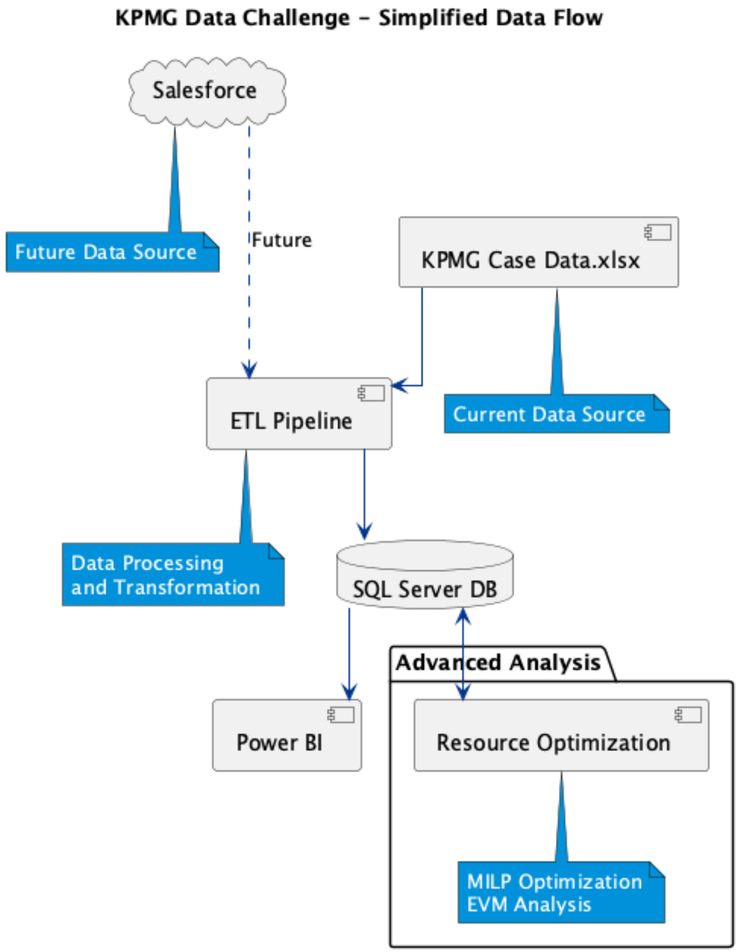
Data Model

ERD Model



Solution Architecture

Technology stack

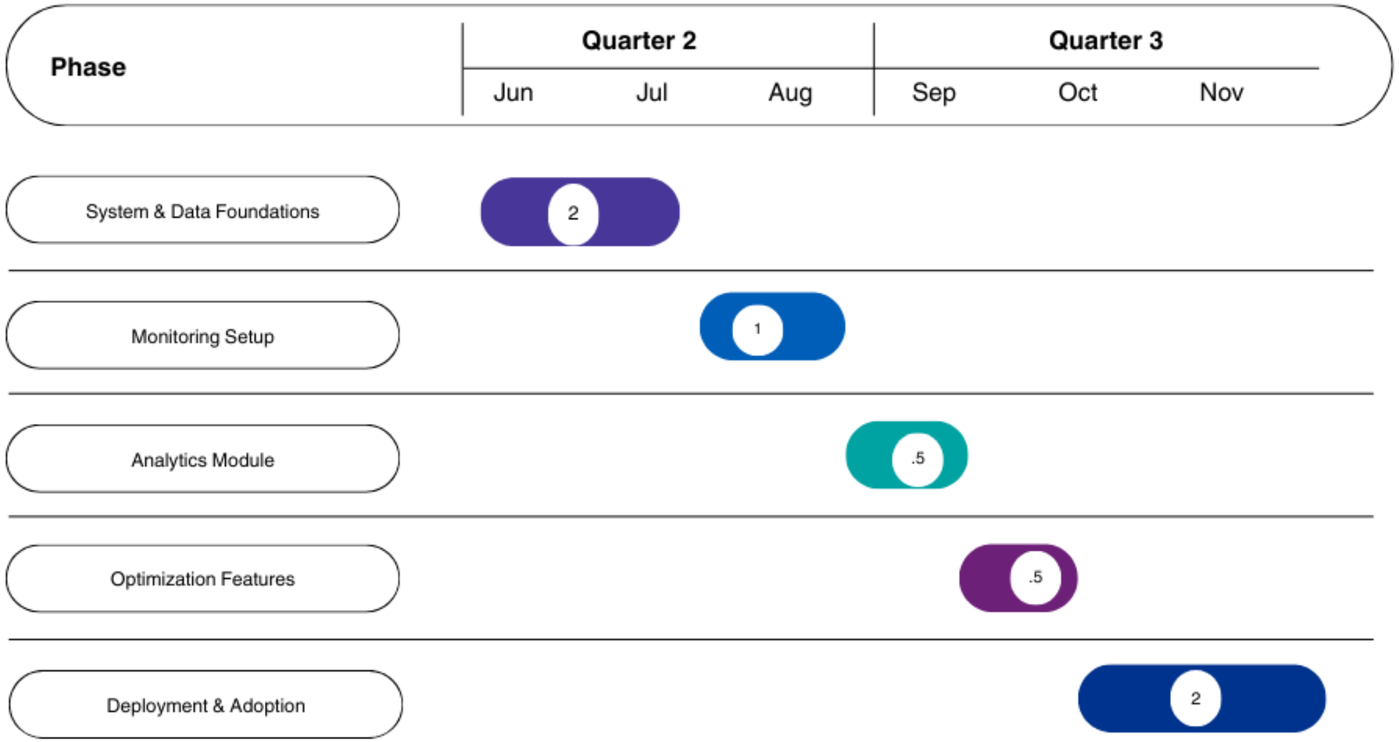


Overview

- **Data Analysis & Optimization:** Python (Pandas, NumPy, PuLP)
- **Data Storage:** SQL Server database (T-SQL), targeting Azure SQL compatibility
- **Visualization:** Power BI
- **Deployment:** Docker in development stage

PROP Implementation Timeline

Gantt chart



All assumptions are discussed in the read.me file

Key Assumptions

And their effect on the model

Linear project progress

CPI

0.98

Efficiency of employees

Charge-out is negotiated per assignment

Impact:

We utilize CPI to estimate the timeline of the project

Remaining assumptions enable the completion of the model

Financial considerations

Spending less time on internal work results in increased bandwidth to charge clients

1

The average manager charge-out rate was at least $250. Each hour saved from staffing analysis and resource allocation can be diverted to charging clients

2

The system lowers bench rates & reduces usage of external consultants, improving profits

Appendix

Please refer to the readme.pdf file for a comprehensive overview of the project

https://github.com/vibqetowi/2025-KPMG-Data-Challenge

Thank you!